

GoGratitude: Using Augmented Reality to Inspire Gratitude

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ABSTRACT

This study explores augmented reality (AR) as a new way to experience gratitude and inspire gratitude in others. We created GoGratitude, an AR application that allows users to create and view digital messages of gratitude connected to a physical location. We tested the application in a usability study ($n = 25$) where we observed participants using the application. Most participants affirmed that this application would help them feel more grateful. Participants also had an overall positive response to the AR experience for expressing gratitude and suggested novel and unexpected uses for the application as well as potential for misuse. We believe this study demonstrates the potential of using AR technology in new ways to positively influence the human-centered design space of gratitude.

CCS CONCEPTS

• **Human-centered computing**; • **Human computer interaction (HCI)**; • **Interaction paradigms**; • **Mixed / augmented reality**;

KEYWORDS

augmented reality, virtual reality, gratitude, unity3d, social interaction

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1 INTRODUCTION

The global COVID-19 pandemic has been a catalyst for creating new opportunities for technology use and the expression of gratitude. Results from a Pew Research Center survey reported 40% of U.S. adults used technology in “new or different” ways because of the pandemic [1]. Additionally, in 2020, many communities participated in a daily ritual of collective outdoor cheering [2] to express their support and gratitude to the community while maintaining

social distancing recommendations. With the pandemic enabling new technology use and increased attention towards gratitude expression, there is a unique opportunity to design new gratitude experiences facilitated by technology.

Technology allows for many pandemic-related advantages for communication. It accommodates social distancing, eliminates physical mediums that transmit disease, and can preserve messages indefinitely. Additionally, technology can provide new ways of sending, receiving, and experiencing digital communication. The benefits of digital communications in a pandemic combined with the possibilities of creating new and interesting ways of expressing gratitude are the motivation for this research. Specifically, this research explores augmented reality (AR) as a new medium for digital messaging focused on gratitude expression.

AR allows users to view digitally created objects within a physical environment through a smartphone, and we explore this as a new communication medium to facilitate, motivate, and preserve expressions of gratitude. GoGratitude is an AR application we designed that lets an individual use a smartphone to create digital messages as AR objects connected to locations in the physical world for other users to view. The interaction is like leaving a physical “sticky-note” at a location except the “sticky-note” exists digitally. This application creates a novel communication experience because it lets users place digital messages for others connected to specific locations, providing an extra level of location-based context. It also changes the way messages are received as they can only be experienced by being physically in the same location where the digital message was created. It can also allow for moments of gratitude to be preserved long-term for others to experience. As an example use case, a university student could leave a digital “sticky-note” outside their teaching assistant’s office to thank them for extra help on a challenging homework assignment. The teaching assistant could receive and view the “sticky-note” through their phone using GoGratitude the next time they came near the message location.

AR has potential to connect digital messages with the physical world and provide new ways to express, experience, and preserve gratitude. The goal of this research is to explore these new interactions.

1.1 Related Work

Research on the science of gratitude and its impact has been broadly studied and documented [3–7]. Recent studies have shown evidence that technology mediated gratitude interventions can have a positive effect on gratitude expression and well-being. These include using technologies such as instant messaging [8], social media [9], and other software developed specifically for gratitude [10, 11].

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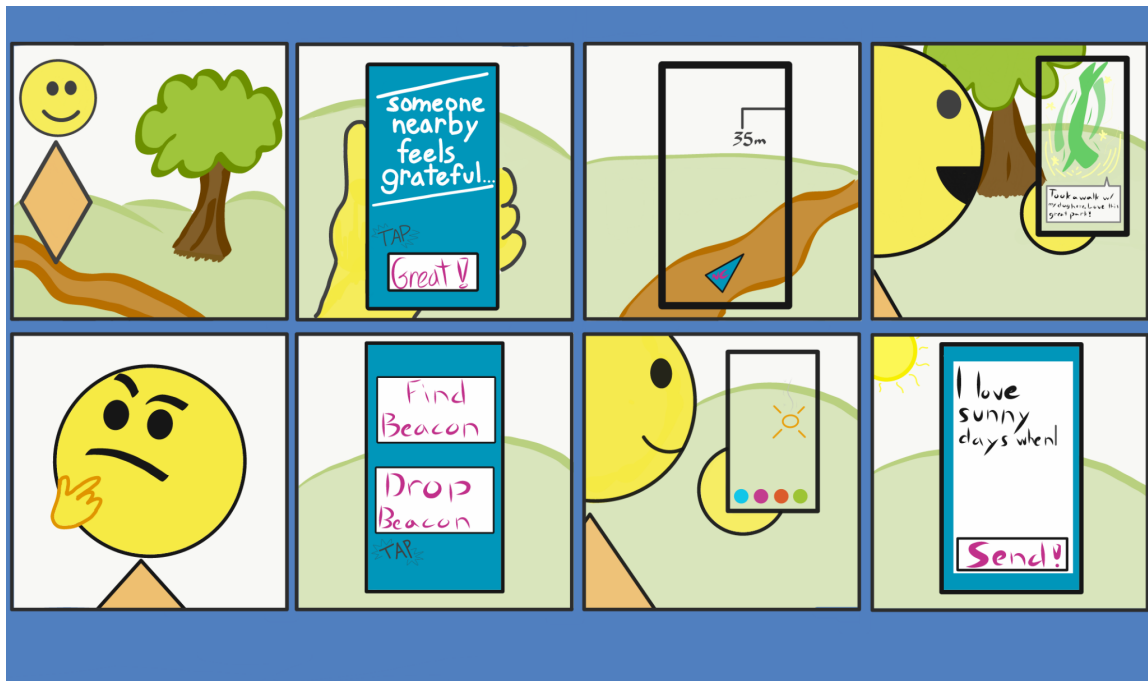


Figure 1: Low-fidelity storyboard prototype

These studies can be considered efforts in a new area of research called “Positive Technology” where novel digital experiences are created for promoting positive change [12, 13]. AR has shown promise as a positive technology for improving behavioral health [14] and as tool for improving communication [15], however, to our knowledge, it has not been used to facilitate gratitude interventions. The pandemic has provided a unique opportunity for technology to spread and encourage communal acts of gratitude. Catungal described the role social media played in Vancouver by encouraging the community to express gratitude for essential workers [16]. Riva et al. considers positive technologies to have the potential to solve pandemic-related problems [17]. We believe exploring AR to inspire gratitude may contribute to new developments in positive technology research.

2 IDEATION AND EXPLORATORY RESEARCH

Gratitude is commonly experienced and expressed in many different forms with direct in-person interactions or by alternative communication methods such as emails, text messages, social media posts, phone calls, letters, etc. These gratitude interactions can be meaningful but our fast-paced technology-enhanced society has been conditioned for digital communication experiences that are fleeting and commonplace. Inspired by the outpouring of public gratitude expressions during the global pandemic, our team decided to create a new way to preserve gratitude expressions using technology. This would then allow others to digitally see and take inspiration from expressions of gratitude which may in turn encourage more acts of gratitude. Motivated by this theme of preserving and sharing moments of gratitude with technology, we explored several ideas. One

of our initial ideas included a simple interactive web-based map application for displaying individual gratitude expressions around the world. Alternatively, more complex physical technologies were considered such as creating an internet-of-things (IoT) companion device dedicated to displaying gratitude messaging between different users. We ultimately decided to explore opportunities that blended digital and physical interactions by using augmented reality technology with smartphone devices. The benefit of using AR on a smartphone is that it combines the ease and scalability of digital technology for preserving gratitude expressions while simultaneously enhancing the digital experience with context and new interactions made possible by a user’s physical location. We decided to create a gratitude messaging application using AR objects that would allow users to create digital messages to not only send and receive location-based expressions of gratitude, but also to motivate and inspire others to express and experience gratitude in new ways.

3 PROTOTYPING

To test initial ideas for GoGratitude and get feedback from potential users, low-fidelity prototypes were used to understand the user experience for gratitude messages in an AR enhanced environment. Figure 1 shows an early storyboard of how the AR interaction would work for creating, viewing, and sharing gratitude messages in the application.

When prototyping, we imagined AR objects with gratitude messages connected to them that could be seen using the application. New messages of gratitude could also be created at a user’s location that could then be viewed by others. We chose to visually represent the gratitude “sticky-note” locations as a beacon that could be seen

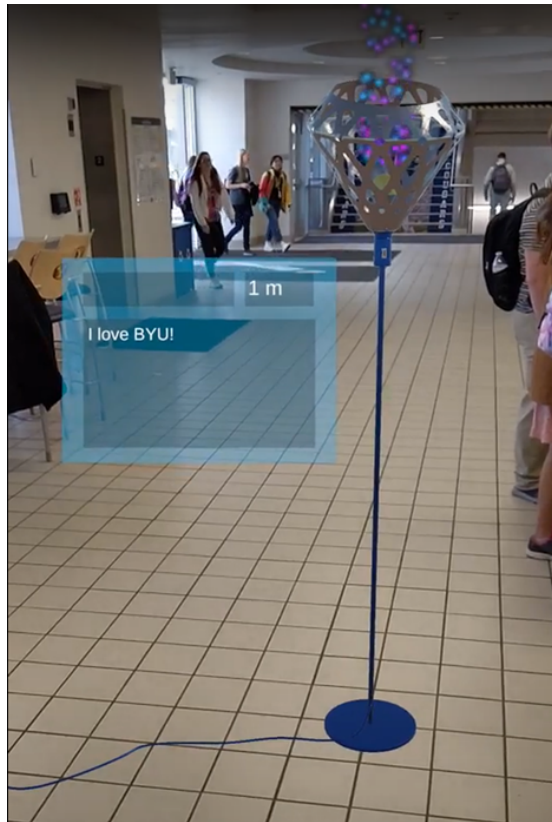


Figure 2: GoGratitude beacon

from a distance and that would get larger as the user gets closer to it. Attached to these beacons would be a message left by the user who placed that beacon. A user could also place their own beacon and message that others could see.

Our first iteration of the application was a functional prototype developed using Unity3D that runs on a smartphone device. It included the ability to place a beacon which would “hook” to the location it was placed. This hook was saved internally to the device. The initial beacon was a tall column of blue light that extended from the source location where it was placed with a note hovering by its side. This beacon of light proved to be difficult to view in an AR environment so it was changed to a lamp-shaped object as shown in Figure 2.

The note hovering by its side remained the same with the addition of a name field to hold information of who posted it. We also created pre-generated beacons so that the first users who opened the application would have beacons to interact with. This would provide data on how users would interact with beacons they did not place. We used this final iteration in our usability testing.

While soliciting input and feedback from other student colleagues during the development of this application (this project was developed as part of an advanced HCI course), it was apparent that this application could be used for creating any type of message, not just expressions of gratitude. Initial prototypes had not implemented a way to prevent users from creating a message

that contained inappropriate or malicious content. As one way to mitigate this kind of misuse, we implemented a basic profanity filter in the application prior to usability testing.

4 USABILITY TESTING METHODOLOGY

We chose to focus the first version of GoGratitude on supporting an on-campus university context, as this type of location facilitates a lot of social interaction among a generally technology savvy community. Study participants were solicited from the Brigham Young University (BYU) community through voluntary response invitations at the Ernest L. Wilkinson Student Center during the mid-day lunch hour. This location is a common gathering area for visitors and students at BYU and had an adequate area for participants to move around and place messages using AR indoors.

At the start of each test, we collected information about a participant’s propensity for expressing gratitude, social media behavior, and demographics (age and gender). Participant’s propensity for expressing gratitude was measured using a 6-question Gratitude Questionnaire (GQ-6) that assesses a person’s disposition towards gratitude [18]. Lower GQ-6 scores indicate a lower disposition towards gratitude while higher scores indicate a higher disposition. Understanding the gratitude disposition of the study participants was important to document because positively or negatively skewed feedback of an application focusing on gratitude expression may correlate with a participant’s inherent disposition towards gratitude. A participant’s social media behavior was also used as an indicator of each participant’s comfort level in using digital platforms for communication. If a participant rarely used social media, for example, it could indicate a participant’s overall concern or negative impression about digital expression and communication which could influence their experience and feedback about GoGratitude. Social media behavior was measured by asking each participant two questions: (1) how often do you use social media and (2) how often do you post on social media? A multiple-choice response was provided for each question ranging from a few times a day to less than a few times a year.

After the initial survey, participants were invited to experience GoGratitude using a smartphone with the application preloaded on it. They were asked to use the application to explore gratitude messages that others had shared around the student center. Additionally, participants were invited to create their own gratitude message and share it in the application.

Once participants finished using the application, they were asked to provide feedback through a semi-structured interview to better understand their experience, how they would use the application, and features they would like to see implemented. We also asked participants whether they felt the application could help them feel more grateful. This usability study was approved by the BYU Institutional Review Board and followed ethical standards and regulations governing human subject research such as having informed consent, using anonymous participant identification, and treating participants fairly and with respect.

We had 25 participants (10 female, 14 male, 1 non-binary) in our usability tests with an age range of 18 to 34 years old (average age = 21.8 years). The majority of participants were current students.

Table 1: Social media behavior from survey responses for the study participants

Social Media Questions	A few times a day	A few times a week	A few times a month	A few times a year	Less than a few times a year
How often do you use social media?	16	5	3	1	0
How often do you post on social media?	0	1	9	10	5

The time spent participating in the study varied among participants but had a range of approximately 5 to 20 minutes.

5 USABILITY TESTING RESULTS

5.1 Propensity for Expressing Gratitude

The GQ-6 scores that measured the participants' propensity towards gratitude ranged from 3.7 to 6.7 (*mean* = 6.0, *median* = 6.2, *SD* = 0.69, *Cronbach's* α = 0.77). Most responses showed a high propensity towards gratitude. This result may have been influenced with an in-person survey response bias. It is likely that asking participants about a personal topic such as their gratitude disposition would guide participants to answer more positively than they otherwise would as to not appear ungrateful to the research team. However, we find this metric to still be useful for determining initial gratitude dispositions of the participants.

5.2 Social Media Behavior

Table 1 shows the distribution of responses for social media behavior of the study participants regarding the two questions posed in the survey. Most participants used social media "a few times a day" while many posted to social media regularly multiple times a month or year. This indicates that most participants in this study commonly used a digital platform to view content and express themselves. From this result, we assume that the study participants were comfortable using digital platforms for communication and that negative feedback in our study is not from a participant's negative impression or lower comfort level towards social media in general.

5.3 Semi-structured Interviews

From the interviews, initial impressions of the GoGratitude application were positive as indicated by 18 out of 25 participants expressing reactions such as, "cool," "fun," "creative," or "great idea." The other 7 participants had varied responses that pointed out application features they liked with feedback such as, "I like it's simplicity," "it's straight to the point," or "it's easy." One participant upon seeing a previous digital message in the application from another participant expressed it was interesting "that someone else was here" and left a positive message they could see. When asked in general what participants thought of the experience, responses also varied, but common themes related to positive feedback such as, "it was pretty fun," "it was cool," "good reminder about gratitude," or "cool to see."

Participants imagined novel uses for the application such as to create "scavenger hunts" or to engage in an "uplifting walk." Others thought it could be used in activities like "geocaching," "treasure

hunting," "setting goals," or going on a "date" using messages along the way. Many of these uses were surprising to us as we had primarily envisioned the application as a way to express gratitude. These creative uses demonstrate how the application could be used in many different contexts and for different purposes. Future application features that were suggested were "reminders," "notifications," and "more accurate placement" of the notes. Other user interface feedback provided suggestions for user interface design elements such as button placement and interactions with the AR elements and are being considered for future iterations.

Finally, we asked participants what they would express gratitude for when using the application and whether it would help them feel more grateful. Some common themes for which our participants would express gratitude included "family," "friends," "health," "BYU," "nature," "environment," "surroundings," and "beauty." The majority of participants (21 out of 25) affirmed that this application would help them feel more grateful. Alternatively, two participants indicated that this may not help them feel more grateful. Two other participants did not answer affirmatively that this application would help them feel more grateful but indicated that the application had the "potential" to help them feel more grateful or "to reflect" about "others around [them]."

5.4 Observations

Due to the interactive nature of AR, we observed that participants wanted to interact more with the AR objects rather than just look at them, such as clicking, liking, and upvoting gratitude messages. Participants were comfortable leaving the immediate vicinity where the study was held to explore the surroundings with the smartphone. Most participants found the AR experience interactive and engaging and spent an average of 3 minutes using the application to view and place a message. One participant was particularly excited about the application and spent approximately 15 minutes walking around with the application and leaving messages around the student center. There was some concern that he may have stolen the mobile device with the application on it and would not return!

A few participants encountered an unexpected challenge when they used the application in a busy hallway of the student center crowded with many other people passing by. These participants expressed reluctance to hold up the smartphone to view AR objects in close proximity to other people because they feared that it would appear that they were recording video or taking pictures of others. Another unexpected observation was that study participants showed excitement for sharing their own ideas about how this application could be used for new exploratory activities as described

previously such as “hide and seek,” “geocaching,” or “treasure hunting.”

Lastly, another interesting aspect of the AR experience was that participants seemed generally familiar with how AR was to be used by holding up the smartphone to view digital content projected into the real world while walking around the surrounding environment. Several participants indicated that they were not familiar with AR technology but they were able to quickly learn how to use the application.

6 IMPROVEMENTS AND FUTURE WORK

Through the design process, we discovered several ways to improve the user experience of GoGratitude. For example, participants suggested adding a “like” or “upvote” feature to the digital notes so that users could do more than just view the digital messages in AR. Several participants recommended a sharing feature where other users could be notified or sent messages directly. The ability to interact with and respond to messages was also suggested as a feature that could improve the interactivity of this experience. Lastly, participants offered several user interface improvements such as making the input buttons and message placement less obstructive.

The topic of misuse of the application, such as creating messages that are inappropriate or malicious, was brought up by a few participants during testing as well as student colleagues during our early investigations. We recognized during application development that filtering user input was going to be challenging beyond a profanity filter, so we chose to focus on the positive use cases of this technology for this study. Future work addressing inappropriate content might use natural language processing to determine and filter on the sentiment or intent of user content or we might explore restricting user input to content appropriate templates that focus on gratitude. We acknowledge that more investigation is needed to address various use cases, including how to appropriately address misuse, to ultimately create the best possible experience for future users.

6.1 Limitations

This research is a first step towards developing and exploring AR technology in new ways for gratitude. As such, this was not a long-term study, yet we feel it still demonstrates valuable insight into the potential of AR as a new medium to experience and facilitate expressions of gratitude. To better understand the impact and influence of AR-based gratitude messaging for users, GoGratitude would need to be tested over an extended period, allowing participants the opportunity for multiple interactions with the application rather than one single experience. This would provide more data and feedback about user engagement, experience, retention, and the impact on the user’s levels of gratitude.

During this study, we encountered unique limitations surrounding research involving gratitude. For example, we found the topic of gratitude to be inherently challenging to study without user bias. Asking participants about their propensity towards gratitude may have encouraged individuals to answer more positively since appearing to be ungrateful is generally not seen as socially acceptable. Another factor that may have affected the levels of gratitude that participants reported was that the study was conducted around the

Thanksgiving holiday in the United States, which may have influenced participants to be more mindful of gratitude in their lives. Even with a likely bias towards a higher propensity for gratitude in this participant sample, we believe that there is still at least preliminary evidence that AR technology has the potential to improve gratitude in users of the GoGratitude application. We also limited our initial investigation to supporting a university community. Based on these limitations, future research should seek to expand the scope of this work to other contexts and user populations.

7 CONCLUSION

In this research study, we explored new gratitude-related experiences using an AR application we developed named GoGratitude. Overall, this new application was received positively among the tested population at BYU and shows potential to create unique ways to motivate and facilitate new opportunities to experience gratitude with those in a university campus community. Most study participants affirmed that this application would help them feel more grateful or would have the potential to help them feel more grateful. Through this study, we have shown the potential of using technology in new ways to positively influence the human-centered design space of gratitude.

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